Fefferman Statement (B) on $\mathbb{T}^3$: A Precise Lean Formalization, a Kernel-Checked Reduction, and the Boundaries of the Frohmanian Symplectic Tether

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Abstract

This monograph records, in Lean 4 against Mathlib, a precise formal statement of Fefferman's Statement (B) on global smoothness of the unforced incompressible Navier–Stokes equations on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, together with an exact ledger of the analytic inputs that the Frohmanian symplectic tether does *not* supply. Statement (B) is *not* proved here. What is proved is a reduction: Statement (B) follows from the classical local theory with the Beale–Kato–Majda continuation criterion and from the open hypothesis that the Beale–Kato–Majda integral is finite on every finite interval of existence. We also record the two-dimensional clock test: constant vortex direction ($\sin\varphi \equiv 0$) coexists with linear growth $\|\phi(t)\|_{L^\infty} = M_0^2\, t$ on steady Euler equilibria.

1 Scope and non-claim boundary

This repository is a boundary and gap ledger. It is **not** a resolution of the Clay Millennium Problem. Every open input is a named proposition in Lean; none is assumed as an axiom, and no proof uses `sorry`. The build and axiom audit (`lean/AxiomReport.lean`, output in `AXIOMS.md`) reports only `propext`, `Classical.choice` and `Quot.sound`. The active continuation of this programme, in which the remaining gap is decomposed into finer named statements (all open or conditional, apart from individually proved lemmas such as the planar `sinShear` obstruction; the open statement is the Frohmanian Tether Hypothesis (A)/(B), see `NAMING.md`), is `BenFrohman/periodic-ns-classical-continuation`.

- **Repository:** `https://github.com/BenFrohman/Clay-`
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2 Statement (B)

Following the official problem description of Fefferman [1], fix $\nu > 0$ and let u° be a smooth, divergence-free vector field on $\mathbb{R}^3$ with $u^\circ(x + e_j) = u^\circ(x)$ for $j = 1, 2, 3$. Take $f \equiv 0$.

> **Statement (B).** There exist smooth functions $p(x,t)$, $u_i(x,t)$ on $\mathbb{R}^3 \times [0,\infty)$ such that
>
> $$\partial_t u_i + \sum_{j=1}^{3} u_j\, \partial_j u_i = \nu\, \Delta u_i - \partial_i p, \qquad \operatorname{div} u = 0, \qquad u(x,0) = u^\circ(x),$$
>
> and u, p are 1-periodic in each spatial coordinate.

In Lean this is the proposition `Clay.StatementB.FeffermanStatementB` (file `lean/Clay/StatementB.lean`). Space is $\mathbb{R}^3$ realised as Euclidean space on Fin 3; partial derivatives are Fréchet derivatives along the standard basis; the time derivative is taken within $[0,\infty)$; smoothness is C^∞ on $[0,\infty) \times \mathbb{R}^3$. No component of the statement is opaque or trivial.

3 The proved reduction

Write $\omega = \operatorname{curl} u$. Consider the propositions

- `PeriodicMaximalSolutionWithBKM` (classical; Kato, Fujita–Kato, Beale–Kato–Majda [2]; not yet formalized): for all admissible data there are $T^* \in (0,\infty]$ and a smooth solution on $[0,T^*)$ such that $T^* < \infty$ implies $\int_0^{T^*} \|\omega(t)\|_{L^\infty}\, dt = \infty$;
- `PeriodicBKMIntegrability` (**open**; the Clay core): every smooth solution on a finite interval $[0,T)$ satisfies $\int_0^{T} \|\omega(t)\|_{L^\infty}\, dt < \infty$.

Theorem 1 (`feffermanStatementB_of_bkm`)**.** `PeriodicMaximalSolutionWithBKM` *and* `PeriodicBKMIntegrability` *together imply Statement (B).* [Proved as an implication; Conditional on both hypotheses]

```
theorem feffermanStatementB_of_bkm (hloc : PeriodicMaximalSolutionWithBKM)
    (hbkm : PeriodicBKMIntegrability) : FeffermanStatementB
```

Theorem 2 (`bkmIntegralFinite_of_holographic`)**.** *If* $\|\omega(t)\|_{L^\infty}$ *is bounded on every finite interval of existence, then* `PeriodicBKMIntegrability` *holds.* [Proved as an implication; the bound itself is open]

The second hypothesis of the first theorem is open and is equivalent in strength to Statement (B) itself. Consequently the reduction does not prove Statement (B).

4 Ledger of missing inputs

The file `lean/Clay/Gaps/MissingInputs.lean` states each former placeholder as a precise proposition:

- `metriplecticControlsEnstrophyL2`: $\sup_{[0,T)} \int_{\mathbb{T}^3} |\omega|^2 < \infty$ for every solution on a finite interval (open);
- `czGivesBMONotLinf`: no constant C gives $\|\nabla v\|_{L^\infty} \le C \|\operatorname{curl} v\|_{L^\infty}$ for smooth periodic divergence-free v (classical, not formalized);

- `spikeObstruction`: no constant C gives $\|f\|_{L^\infty} \le C\|f\|_{L^6(\mathbb{T}^3)}$ for smooth periodic f (classical, not formalized);
- `holographicNotAZFCBound`: the uniform ceiling $\sup_{[0,T)} \|\omega\|_{L^\infty} < \infty$ (open);
- `depletedStretching`: $\omega \cdot S\omega \le C|\omega|^2$ uniformly on finite intervals (open).

The former trivial definitions have been removed; the repository history retains them. For planar fields $v = (v_1(x_1, x_2), v_2(x_1, x_2), 0)$ the vorticity is proved parallel to e_3. Kernel-checked elementary results include the identity $\Pi_u u = 0$ for the pointwise tether projection, the unboundedness of the history integral $m^2 t$ for $m > 0$, and the conditional Riccati rearrangement.

5 Build and axiom status

Proved (kernel-checked). `lake build` of the full `Clay` library (root module and all 13 submodules) succeeds on Lean/Mathlib v4.16.0. The axiom report lists 60 theorem declarations (including compiler-generated auxiliary lemmas), none using `sorryAx`, and none using an axiom beyond `propext`, `Classical.choice`, `Quot.sound`. The substantive kernel-checked results are `feffermanStatementB_of_bkm`, `statementB_of_bkmIntegralFinite`, `bkmIntegralFinite_of_holographic`, `bkmIntegralFinite_of_vortSup_le`, the two-dimensional clock results `steady_euler_clock`, `alignment_does_not_delete_clock`, `clock_fails_on_aligned_2d_solutions`, the planar-vorticity lemma `curl_planar`, the tether identities `tether_degeneracy`, `tether_bracket_preservation`, and the integer rearrangement `conditional_riccati`. This is a machine check of the Lean statements, not peer review.

Conditional on. `PeriodicMaximalSolutionWithBKM` (classical, not formalized here), `PeriodicBKMIntegrability` (open), and, for `conditional_riccati`, `CorrectedMaxRate`, an *open hypothesis* and not a classical result: it is a Riccati / κ-matching max-rate law $\dot{M} \le CM - \kappa M^2$ at the vorticity maximum. The argument proposed for it is refuted, since the stretching at the maximum is quadratic, $\omega \cdot S\omega \le C_{CZ} M^2$ (up to a logarithm), and nothing in unmodified Navier–Stokes supplies a matching $-\kappa M^2$ term; `conditional_riccati` only rearranges this hypothesis. Finer decompositions of the open input (for example tame H^s and log-Sobolev BKM bounds) belong to the continuation repository and are not formalized here.

Open. `PeriodicBKMIntegrability`; every item of the ledger above marked open; and therefore Statement (B) itself. The Clay Millennium Problem is not solved by this work.

6 Provenance hashes

Source files are fingerprinted with SHA-256 (`provenance/PROVENANCE_HASHES.json`) and a Merkle root (`provenance/PROVENANCE_MERKLE_ROOT.json`), signed with Ed25519 key fingerprint `b428fb72676f1c52`. These are file-integrity fingerprints of an earlier snapshot only; they are not a certificate of correctness or review. The authoritative correctness check is the build and axiom report above, rerun by CI on every push.

References

[1] C. L. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, in: The Millennium Prize Problems, Clay Mathematics Institute, 2006, pp. 57–67.

[2] J. T. Beale, T. Kato, A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.

[3] P. Constantin, C. Fefferman, *Direction of vorticity and the problem of global regularity for the Navier–Stokes equations*, Indiana Univ. Math. J. **42** (1993), 775–789.

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